

Biashara AI

Mobile App Design Document

Empowering African SMEs with On-Device AI

Version 2.0 — May 2026

1. Introduction and App Concept

Biashara AI (Swahili for "Business AI") is a pioneering mobile application designed to empower Small and Medium Enterprises (SMEs) across Africa. By leveraging Google's Gemma on-device AI models via the LiteRT-LM runtime, the app provides a suite of intelligent business management tools that function entirely offline.

This approach directly addresses the continent's challenges of limited internet connectivity, high data costs, and the need for culturally relevant, local-language solutions. Biashara AI aims to transform how SMEs manage their inventory, sales, and finances, fostering economic growth and digital inclusion.

Important note on device compatibility: On-device LLM inference requires a mid-to-high range device (e.g., Qualcomm Snapdragon 8-series or equivalent). The app includes a device capability check at launch and gracefully disables AI features on unsupported hardware, falling back to rule-based tools.

2. Target Users and Market Context

The primary users of Biashara AI are SME owners and managers in Sub-Saharan Africa, operating in sectors such as retail, services, and small-scale production. These entrepreneurs often face a "digital divide" where existing business software is either too expensive, requires constant internet, or is not available in their native language.

User Segment	Typical Characteristics	Key Pain Points
Market Vendors	High transaction volume, limited storage, often informal.	Stockouts, difficulty tracking daily profits, language barriers.

User Segment	Typical Characteristics	Key Pain Points
Small Shopkeepers	Diverse product range, regular customers, semi-formal.	Inefficient inventory tracking, lack of data for growth, high data costs.
Artisans & Producers	Project-based, material-intensive, niche markets.	Managing material costs, pricing products, finding new customers.
Service Providers	Time-based, recurring customers, low overhead.	Appointment management, tracking service history, financial planning.

3. Core Features and AI Capabilities

Biashara AI integrates essential business functions with the power of on-device Gemma AI via the LiteRT-LM runtime.

3.1 Intelligent Inventory and Sales Management

The app provides a robust system for tracking products and sales. Gemma 3 1B (4-bit quantized, ~600MB) performs on-device demand forecasting by analyzing local sales history, predicting upcoming stock needs, and alerting owners to potential shortages before they occur. This helps SMEs optimize their working capital and reduce waste.

3.1.1 Barcode and QR Code Scanning

Product entry via the phone camera is supported using CameraX and ML Kit Barcode Scanning — entirely offline. Users can scan product barcodes (UPC-A, UPC-E, EAN-13, EAN-8) and QR codes to instantly populate product fields, look up existing inventory, or record a sale. This eliminates manual entry errors and dramatically speeds up stock management for market vendors handling high transaction volumes.

Supported scanning flows:

- Scan to add product: Scans a barcode and auto-fills the product name and barcode fields in the Add Product screen.
- Scan to record sale: Opens the POS screen with the scanned product pre-selected.
- Scan to update stock: Scans a barcode and navigates directly to that product's stock adjustment screen.

3.2 Personalized Customer Insights

A simple Point-of-Sale (POS) interface allows for quick transaction recording. The integrated recommendation engine (powered by a lightweight on-device classifier, supplemented by Gemma for natural language summaries) suggests products based on customer purchase history. All processing occurs on-device, enabling a personalized shopping experience with zero data costs.

3.3 Financial Health and Insights

Biashara AI simplifies expense and revenue tracking. The Cash Flow Insights feature uses Gemma to analyze financial trends and provide actionable advice in simple, everyday language — for example, alerting to an unusual spike in expenses or suggesting ways to improve profitability based on recorded data.

3.4 Local Language and Voice Accessibility

Biashara AI supports major regional languages including Swahili, Hausa, Yoruba, and Amharic. The app features on-device voice interaction, allowing users to manage their business through spoken commands. Amharic support includes full Ethiopic script rendering and right-to-left-aware layout adjustments. Language selection is available at onboarding and can be changed at any time from Settings.

4. User Interface and Experience (UI/UX) Design

The design of Biashara AI prioritizes simplicity, accessibility, and an offline-first philosophy.

4.1 Navigation and Layout

The app uses a bottom navigation bar with five sections: Home (dashboard), Inventory, Sales (POS), Insights, and Settings. This pattern is familiar to users of popular consumer apps and requires minimal digital literacy. All primary actions are reachable within two taps from the home screen.

4.2 Key Screen Flows

Screen	Primary Action	AI Feature
Home Dashboard	View daily summary, low-stock alerts	AI-generated daily tip
Inventory List	Browse products, scan barcode, add item	Demand forecast badge
Add / Edit Product	Enter product details or scan to fill	Voice input for all fields
POS / Sales	Record sale by scan or manual select	Product recommendation
Cash Flow Insights	View income/expense trends	Gemma narrative summary
Settings	Language, sync, AI model status	Model download manager

4.3 Language-First Onboarding

Users select their preferred language immediately upon first launch. The selection screen displays language names in their native script (e.g., "አማርኛ" for Amharic). All subsequent UI text, voice prompts, and AI-generated content is presented in the chosen language.

4.4 Visual Dashboards

Key business metrics are presented through simple charts and summaries, providing at-a-glance insights. Charts use high-contrast colors tested against WCAG AA contrast standards. All text labels are rendered at a minimum 14sp to support users with limited visual acuity.

4.5 Offline-First Indicators

The app functions identically whether online or offline. A subtle status chip in the top bar shows sync status. When connectivity is available, a background sync runs automatically. The model download manager (in Settings) shows the AI model download status and allows re-download if storage was cleared.

4.6 Accessibility

- Minimum 48dp touch targets for all interactive elements.
- Right-to-left layout support for Amharic (Ethiopic script).
- TalkBack / screen reader labels on all icons and charts.
- High-contrast mode option in Settings.

5. Technical Architecture and AI Integration

Biashara AI is built on a modern, robust architecture designed for the unique constraints of the African mobile market.

5.1 On-Device AI Pipeline (LiteRT-LM)

As of 2026, the correct runtime for on-device Gemma on Android is LiteRT-LM, Google's production-ready inference framework (successor to the deprecated MediaPipe LLM Inference Android API). Models are distributed in .litemlm or .task format and downloaded at first launch — they are too large to bundle in an APK.

- Runtime: LiteRT-LM via com.google.mediapipe:tasks-genai (current stable)
- Primary model: Gemma 3 1B, 4-bit quantized (~600MB), downloaded from HuggingFace LiteRT Community on first run
- Fallback: Rule-based tools for devices that fail the minimum hardware check (Gemma 3 1B requires approximately 2GB RAM free)
- Quantization: INT4 to minimize memory and battery usage on mid-range smartphones

- Local Inference: All AI processing occurs on the device CPU/GPU, ensuring zero latency and zero data costs once the model is downloaded

5.2 Barcode Scanning Stack

- Camera: CameraX (androidx.camera:camera-core, camera-camera2, camera-lifecycle, camera-view)
- Barcode detection: Google ML Kit Barcode Scanning (com.google.mlkit:barcode-scanning)
- Integration: MKitAnalyzer bridges CameraX and ML Kit with automatic coordinate transformation
- Supported formats: UPC-A, UPC-E, EAN-13, EAN-8, QR Code — all processed fully on-device with no internet required

5.3 Data and Security

- Local Storage: All business data stored in an encrypted SQLite database via Room Persistence Library
- Privacy-by-Design: Sensitive business information never leaves the user's phone
- Optional Sync: Users may synchronize data with a secure cloud backend for backup and multi-device access when internet is available

5.4 Device Compatibility

On-device LLM inference is computationally demanding. The app performs a capability check at launch using the following criteria:

Tier	Criteria	AI Features Available
Full AI	4GB+ RAM, Android 10+, Qualcomm 800-series or equivalent	All features: demand forecast, voice, insights, recommendations
Partial AI	3GB RAM, Android 9+	Cash flow insights only (lighter prompt)
Rules-based	Below threshold	No LLM features; rule-based alerts and manual entry only

6. Conclusion

Biashara AI represents a significant step toward bridging the digital divide for African SMEs. By bringing the power of advanced AI models directly to the devices people already own — and pairing them with practical tools like barcode scanning — we can provide sophisticated business tools that are affordable, accessible, and culturally relevant.

This design document reflects the current technical landscape as of May 2026, including the migration from deprecated MediaPipe Android APIs to the production LiteRT-LM runtime, and the addition of camera-based barcode scanning for inventory management.

7. References

- [1] Google AI Edge. LLM Inference guide for Android.
https://ai.google.dev/edge/mediapipe/solutions/genai/llm_inference/android
- [2] Google AI Edge. LiteRT-LM GitHub. <https://github.com/google-ai-edge/LiteRT-LM>
- [3] Google ML Kit. Scan barcodes with ML Kit on Android. <https://developers.google.com/ml-kit/vision/barcode-scanning/android>
- [4] Android Developers. ML Kit Analyzer (CameraX).
<https://developer.android.com/media/camera/camerax/mlkitanalyzer>
- [5] LiteRT Community on HuggingFace. <https://huggingface.co/litert-community>
- [6] Crane AI Labs. Swahili Gemma 1B.
<https://deepmind.google/models/gemma/gemmaverse/craneai/>